1 Solution — GIAM 1.6 Problem 5

1.1 Problem

Write a proof that $\sqrt{3}$ is irrational.

1.2 Strategy

We mimic the classical proof for $\sqrt{2}$, using the case $p = 3$ of the lemma from Problem 4:

$$\text{if } 3 \mid x^2, \text{ then } 3 \mid x.$$

We first prove this lemma, then run the contradiction argument.

1.3 The Lemma: $3 \mid x^2 \Rightarrow 3 \mid x$

Every integer x leaves remainder 0, 1, or 2 on division by 3. Squaring each case and reducing mod 3:

$x \equiv \pmod 3$	$x^2 \equiv \pmod 3$	$3 \mid x^2$?
0	0	yes
1	1	no
2	$4 \equiv 1$	no

Table 1.1

A square is divisible by 3 **only** in the first row — precisely when x itself is divisible by 3. Hence $3 \mid x^2 \Rightarrow 3 \mid x$. (Equivalently, this is Euclid's Lemma with the prime 3.) $\square$

1.4 The Proof That $\sqrt{3}$ Is Irrational

The lemma for p = 3: a square is divisible by 3 only when its root is. Check all residues mod 3.

if x ≡ … (mod 3)	then x^2 ≡ … (mod 3)	3 | x^2 ?
0 (so 3 | x)	$0^2 = 0$	yes
1	$1^2 = 1$	no
2	$2^2 = 4 \equiv 1$	no

x^2 is a multiple of 3 only in the top row, where x itself is a multiple of 3. Hence $3 \mid x^2 \Rightarrow 3 \mid x$.

Proof that √3 is irrational (by contradiction)

1. Suppose not: $\sqrt{3} = a/b$ with gcd(a, b) = 1 (lowest terms).
2. Square and clear: $3 = a^2/b^2 \Rightarrow a^2 = 3b^2$. So $3 \mid a^2$.
3. By the lemma, $3 \mid a$. Write $a = 3c$.
4. Substitute: $(3c)^2 = 3b^2 \Rightarrow 9c^2 = 3b^2 \Rightarrow b^2 = 3c^2$. So $3 \mid b^2$.
5. By the lemma again, $3 \mid b$.
6. But then $3 \mid a$ and $3 \mid b$, so gcd(a, b) ≥ 3 — contradicting step 1.

Contradiction ⇒ √3 is irrational. ∎

Figure 1.1: Top: residues mod 3. If $x \equiv 0$ then $x^2 \equiv 0$ (divisible by 3); if $x \equiv 1$ then $x^2 \equiv 1$; if $x \equiv 2$ then $x^2 \equiv 4 \equiv 1$. So x^2 is a multiple of 3 only when x is, giving $3 \mid x^2 \Rightarrow 3 \mid x$. Bottom, the proof by contradiction: (1) suppose $\sqrt{3} = a/b$ with gcd(a,b)=1; (2) then $a^2 = 3b^2$, so $3 \mid a^2$; (3) by the lemma $3 \mid a$, write $a = 3c$; (4) then $9c^2 = 3b^2$, so $b^2 = 3c^2$, so $3 \mid b^2$; (5) by the lemma $3 \mid b$; (6) but then 3 divides both a and b, contradicting gcd(a,b)=1. Therefore $\sqrt{3}$ is irrational.

Theorem. $\sqrt{3}$ is irrational.

Proof. Suppose, for contradiction, that $\sqrt{3}$ is rational. Then we may write it in lowest terms,

$$\sqrt{3} = \frac{a}{b}, \qquad a, b \in \mathbb{Z},\ b \neq 0,\ \gcd(a, b) = 1.$$

Squaring both sides and clearing the denominator:

$$3 = \frac{a^2}{b^2} \quad \Longrightarrow \quad a^2 = 3\,b^2.$$

Thus $3 \mid a^2$, and by the lemma $3 \mid a$. Write $a = 3c$ for some integer c. Substituting,

$$(3c)^2 = 3\,b^2 \quad\Longrightarrow\quad 9c^2 = 3\,b^2 \quad\Longrightarrow\quad b^2 = 3\,c^2.$$

So $3 \mid b^2$, and by the lemma again $3 \mid b$. But now 3 divides *both* a and b, so $\gcd(a,b) \geq 3$ — contradicting $\gcd(a,b) = 1$.

The assumption that $\sqrt{3}$ is rational is therefore false, so $\sqrt{3}$ is irrational. ■

1.5 Remark — Where the Argument Differs From $\sqrt{2}$

Structurally the proof is *identical* to the $\sqrt{2}$ argument — only the number 3 plays the role of 2. The one place care is needed is the lemma. For $\sqrt{2}$ the lemma "$2 \mid x^2 \Rightarrow 2 \mid x$ " is the parity statement of Problem 3, proved by writing an odd $x = 2k+1$. For $\sqrt{3}$ the analogous move is to check the *three* residues mod 3, as in the table: x^2 is never $\equiv 2 \pmod 3$, and is $\equiv 0$ only when $x \equiv 0$. The residue-checking method scales to any prime and is the concrete engine behind the general Problem 4 lemma.

1.6 Remark — Why "Lowest Terms" Is the Trap That Springs

The contradiction is powered entirely by the assumption $\gcd(a,b) = 1$. The algebra forces 3 to divide both a and b, which is impossible for a *reduced* fraction. This "infinite descent in one step" is the heart of the method: if $\sqrt{3} = a/b$ existed, then $a/3$ and $b/3$ would give a *smaller* representation $\sqrt{3} = (a/3)/(b/3)$, and one could repeat forever — impossible for positive integers. Insisting on lowest terms at the outset packages that infinite regress into a single, clean contradiction.

1.7 Remark — The Non-Square Modulus Would Fail

It is worth seeing why the same script does *not* prove "$\sqrt{4}$ is irrational" (it had better not — $\sqrt{4} = 2$). Rerunning with 4: from $a^2 = 4b^2$ we get $4 \mid a^2$, but the lemma "$4 \mid x^2 \Rightarrow 4 \mid x$" is **false** ($x = 2$ gives $4 \mid 4$ yet $4 \nmid 2$). The chain breaks at the first step. This is exactly the point of Problem 4: the argument needs a *prime* (or, more generally, a non-square with some prime to an odd power). Since 3 is prime, the lemma holds and $\sqrt{3}$ is irrational; since $4 = 2^2$ is a perfect square, it isn't.

1.8 Remark — A One-Line Factorization Proof

For those who prefer the prime-factorization viewpoint (Problem 1.5.3), here is the whole argument compressed. In $a^2 = 3b^2$, count the exponent of the prime 3 on each side. On the left, a^2 has an **even** exponent of 3 (squaring doubles it). On the right, $3b^2$ has an **odd** exponent of 3 (an even exponent from b^2, plus one more from the explicit factor 3). An integer cannot have both an even and an odd exponent of 3 in its unique factorization — contradiction. The same parity-of-exponent clash proves $\sqrt{n}$ irrational for every non-square n.